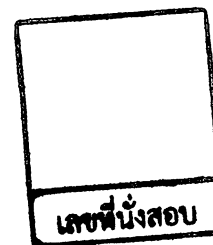


College of Industrial Technology

King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2017

Subject: 392131 Physics I

Section: 15-18

Date: 5 October 2017

Time: 10:00-12:00

Name: _____ ID: _____ Class: _____

Instructions:

1. The examination has 8 pages (including this page) and a total score of 70 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are **NOT** allowed in the examination.
9. Electronic communication devices are **NOT** allowed in the examination room.
10. If not specified otherwise, used these values: $g = 10 \text{ m/s}^2$, $\pi = 3.14$.

**Cheating in the exam is considered an extremely serious
offence which will result in expulsion from the University**

(1) Convert the following to SI Units. Work across the line and show all steps in the conversion. (Given: 1 in = 2.54 cm, 1 liter = 1,000 cm³) (5 marks)

- a. 72 km/hr =
- b. 3 liter/min =
- c. 12 in² =
- d. 250 cm³ =
- e. 216 km/min² =

(2) Consider the equation

$$y = x + \frac{1}{2} kx^3$$

If the above equation x and y are both lengths (dimension, [L]). What is the dimension of k? (5 marks)

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(3) Compute the following numbers, applying the significant figure standards adopted for this text. (5 marks)

- a. $11.1 \times 25.4 = \dots\dots\dots$
- b. $111.11 - 25.4 = \dots\dots\dots$
- c. $63.50 + 25.4 = \dots\dots\dots$
- d. $1250/25.4 = \dots\dots\dots$
- e. $25.4^2 = \dots\dots\dots$

(4) A car travelling initially at 50.0 ± 0.2 m/s undergoes constant acceleration of 2.00 ± 0.01 m/s² for a time of 4.0 ± 0.1 seconds. From the above values and its uncertainty, calculate the total displacement of this car. (5 marks)

Note: Both significant figure and uncertainty analysis are required in this problem and the

displacement $s = ut + \frac{1}{2}at^2$

(5) A raven flies to the South for 120 seconds with speed of 12 m/s, then flies to the North 0.56 km for 80 seconds. Find the total distance, total displacement, average speed and average velocity. (4 marks)

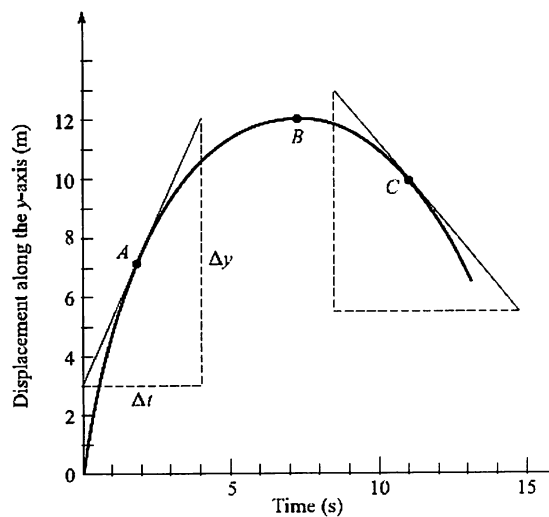
(6) A man starts from home by walking 4 m/s in 4 seconds and then running by increasing velocity 4 m/s each 2 second. The total time used for running is 10 seconds. After that, the man slows down by 4 m/s^2 until stop the motion.

a. Sketch the graph of velocity-time to explain moving of this man. (3 marks)

b. Calculate total displacement of the man from the above graph. (3 marks)

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(7) The vertical motion of an object shown in the figure. Find its instantaneous velocity at points A, B and C (6 marks)



(8) A bus moving at a speed of 20 m/s begins to slow at a constant rate of 2 m/s each second until it stops. Find:

- a. How far it goes before stopping? (3 marks)
- b. How long does a bus move? (3 marks)

(9) A jet plane lands with a speed of 100 m/s and can accelerate at a maximum rate of 25 m/s^2 as it comes to a rest.

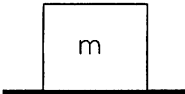
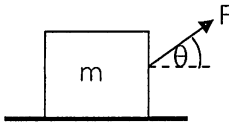
- a. From the instant the plane touches the runway, what is the minimum time needed before it can come to a rest? (3 marks)
- b. Can this plane land on a small tropical island airport where the runway is 0.80 km long? (3 marks)

(10) A rocket rises vertically, from rest, with an acceleration of 4 m/s^2 until it runs out of fuel at an altitude of 800 m. After this point, its acceleration is that of gravity, downward.

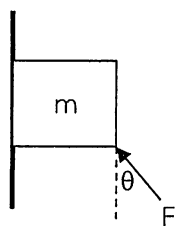
- What is the rocket reach? (3 marks)
- How long (total) is it in the air? (3 marks)

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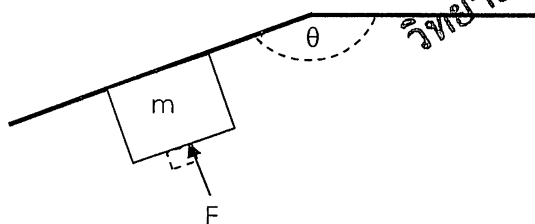
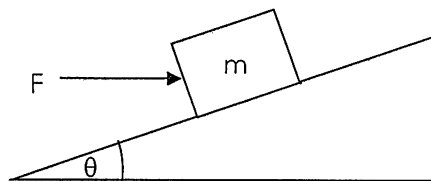
(11) For each figure, write the free body diagram and find the value of normal force (N) in term of m , g , θ and F . (No friction force) (10 marks)

(a) 	(b) 
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(c)



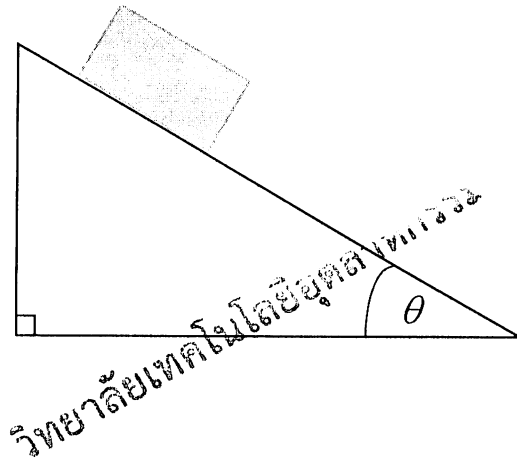
(d)



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(12) A carton shown in the figure lies on a plane tilted at an angle $\theta = 37^\circ$ to the horizontal with $\mu_k = 0.5$. Solve this problem from these questions as below:

- Determine the acceleration of the carton as it slides down the plan (3 marks)
- If the carton starts from rest 6 m up the plane from its base, what will be the carton's speed when it reaches the bottom of the incline? (3 marks)



Instructor team

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